

2015 onward

2015

Faster gene editing

The CRISPR-Cas9 tool was developed by US researchers to allow scientists to edit genes of living things and swap parts of a genome (complete set of inherited information stored in each of an organism's cells) in and out faster and more accurately than before. Using CRISPR-Cas9, scientists developed a mosquito that was resistant to catching the parasite that causes the disease malaria.

2015

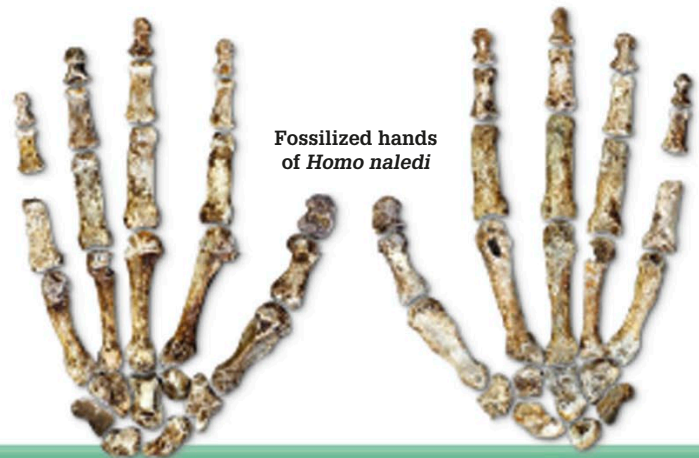
New human ancestor

A new hominid (early human) species was described and named *Homo naledi*. Extensive fossil remains of this previously unknown human ancestor were found in the dolomite rock of the Rising Star cave system in South Africa in 2013. Around 1,500 fossil fragments of some 15 different individuals were recovered. This human ancestor stood an estimated 59-in (150-cm) tall, walked on two legs, and is thought to have lived about two million years ago.

IVF pups

In 2015, seven puppies became the first "test-tube" dogs to be born as a result of IVF (in vitro fertilization), in which male sperm cells fertilize female egg cells in a laboratory. Other species may be created in the same way.

One of the seven puppies born after IVF research at Cornell University, New York



Fossilized hands of *Homo naledi*



2015

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First photos of Pluto

After a nine-year journey, NASA's New Horizons space probe finally reached the dwarf planet Pluto. It sent back the first detailed images of Pluto and its moon, Charon. They revealed a more varied landscape than expected, including mountain ranges, ice volcanoes, dunes, and nitrogen ice fields.



Pluto as seen by the New Horizons space probe

2015

Evidence of water on Mars

NASA announced evidence that liquid water may have flowed on Mars under present conditions. This stemmed from findings by NASA's Mars Reconnaissance Orbiter probe that revealed streaks on some slopes on the Martian surface, which may have been made by salty liquid water.



Mars Reconnaissance Orbiter

2015

New antibiotic

Teixobactin, the first new type of antibiotic in more than 20 years, was recovered from soil bacteria. A team at Northeastern University in Boston, MA, used an electronic chip to grow microbes in soil and then isolated their antibiotic chemical compounds.